

KAIRI-SAFE Risk Measurement and Applications to Financial Model Risk Assessment Framework

Binhe Jia

¹Department of Mathematical Science, University of Toronto

²Toronto, Canada

`binhe.jia@mail.utoronto.ca`

Abstract

The increasing adoption of artificial intelligence in financial services has created new challenges for model risk management. Although machine learning models can improve predictive performance in applications such as credit scoring, portfolio management, fraud detection, and risk forecasting, they also introduce risks related to robustness, accuracy, fairness, and explainability. This paper builds on the KAIRI-SAFE framework by examining how Sustainability, Accuracy, Fairness, and Explainability can be measured using statistical indicators and integrated into financial risk management processes. Statistical tools including the Diebold–Mariano test, DeLong test, Kolmogorov–Smirnov test, t-test, Shapley values, and Gini coefficients are introduced as methods for evaluating AI model risk. The paper further maps SAFE indicators to financial risk categories such as model risk, market and credit risk, regulatory risk, and governance risk. Finally, an aggregate SAFE rating is proposed to combine individual indicators into a model-level risk assessment while preventing severe weaknesses in one dimension from being offset by strong performance in another. The proposed framework provides a structured approach for translating AI model evaluation into operational risk indicators that can support governance, monitoring, and decision-making in financial institutions. An illustrative credit scoring example demonstrates how the aggregate SAFE rating can translate statistical diagnostics into concrete model risk management actions.

Keywords: Artificial intelligence, model risk management, SAFE framework, explainable AI, financial risk, machine learning governance

1 Introduction

In the past several years, we have witnessed a boom of AI applications in payments, asset management, and insurance. While AI systems allow firms and investors to analyze complex financial datasets and execute trades at much higher speeds, they pose significant challenges for risk management. First, unlike traditional financial models which rely solely on financial features such as interest, asset price, volatility, AI models often rely on features extracted from datasets through pretraining, which inherently lack direct connection to financial context. Second, deep learning models such as BERT and GNN are known to be "black-box" and the training process of the model is often difficult to trace, which becomes a significant source of Knightian uncertainty for model risk management (Board of Governors of the Federal Reserve System and Office of the Comptroller of the Currency, 2011; National Institute of Standards and Technology, 2023).

While AI model risk is still actively researched, many measurements have been proposed. Bracke et al. (2019) apply explainability methods to default risk analysis, while Lundberg and Lee (2017) propose SHAP, a Shapley-value-based approach for explaining model predictions.

This paper builds on Giudici, Centurelli, and Turchetta (2024), who propose a series of quantitative key risk indicators for AI: Sustainability (S), Accuracy (A), Fairness (F), and Explainability (E). This paper discusses the intuition behind each indicator and its implications for financial model and operational risk. Finally, this paper proposes a framework for managing AI-related financial risk based on these implications, with alignment to Basel model risk guidance, the NIST AI Risk Management Framework, ISO/IEC 23894:2023, and the EU AI Act (Board of Governors of the Federal Reserve System and Office of the Comptroller of the Currency, 2011; European Parliament and Council of the European Union, 2024; International Organization for Standardization, 2023; National Institute of Standards and Technology, 2023).

2 Statistical Tools for SAFE Measures

In order to evaluate the SAFE metrics and compare machine learning models, several statistical hypothesis tests are employed. These tests are used to compare prediction errors, classification performance, distributions, and parameter significance. The main statistical tests used in this work are the Diebold–Mariano test, the DeLong test, the Kolmogorov–Smirnov test, the t-test, and the χ^2 test. Additionally, key indicators (Shapley value, Gini coefficient) are introduced in this section.

2.1 Diebold-Mariano Test

Diebold–Mariano Test was originally developed to compare the forecast accuracy of two time series models (Diebold and Mariano, 1995).

Define d_t as the loss differential between two forecasting models $\hat{y}_{1,t}$ and $\hat{y}_{2,t}$. A hypothesis testing is conducted using the following hypothesis:

- $H_0 : \mathbb{E}[d_t] = 0$
- $H_a : \mathbb{E}[d_t] \neq 0$

Define the model errors:

- $e_{1,t} = y_t - \hat{y}_{1,t}$
- $e_{2,t} = y_t - \hat{y}_{2,t}$

Then, a loss function L is defined. This can be e^2 , $|e|$, or any custom loss functions which does not have to be symmetric or quadratic. Using L , we define $d_t = L(e_{1,t}) - L(e_{2,t})$ and:

- If $d_t > 0$, we conclude that model $\hat{y}_{2,t}$ is better.
- If $d_t < 0$, we conclude that model $\hat{y}_{1,t}$ is better.

Define the mean loss differential:

$$\bar{d} = \frac{1}{T} \sum_{t=1}^T d_t$$

Define Diebold-Mariano test statistic:

$$DM = \frac{\bar{d}}{\sqrt{\text{Var}(\bar{d})/T}}$$

Under the H_0 , we have $DM \sim N(0, 1)$ and :

- $DM \approx 0$: Models have similar accuracy

- $DM > 0$: Model $\hat{y}_{2,t}$ has better accuracy
- $DM < 0$: Model $\hat{y}_{1,t}$ has better accuracy

Additionally, if the p-value $p < 0.05$, we may conclude that such difference is statistically significant.

2.2 DeLong Test

The DeLong test is a statistical method used to compare the AUCs of correlated ROC curves (E. R. DeLong, D. M. DeLong, and Clarke-Pearson, 1988). The test is based on U-statistics and estimates the variance of the difference between the areas under the ROC curves (AUCs) by decomposing the AUC into contributions from positive and negative samples. The statistical significance of the difference between two classifiers is then assessed using a normal approximation.

The receiver operating characteristic (ROC) curve is a widely used tool for evaluating binary classifiers. It is a plot of the true positive rate (TPR) against the false positive rate (FPR) for different classification thresholds:

$$TPR = \frac{TP}{TP + FN}$$

$$FPR = \frac{FP}{FP + TN}$$

where

1. **TP**: True positive, which is the number of result with both prediction and target being positive.
2. **FN**: False negative, which is the number of result with negative prediction but positive target.
3. **TN**: True negative, which is the number of result with both prediction and target being negative.
4. **FP**: False positive, which is the number of result with positive prediction but negative target.

The area under the ROC curve (AUC) provides a scalar measure of classification performance. The DeLong test aims at testing the following hypothesis:

$$H_0 : AUC_1 = AUC_2$$

where

$$AUC = \mathbb{P}(\text{positive} > \text{negative}) = \frac{1}{mn} \sum_{i=1}^m \sum_{j=1}^n \phi(s_i^+, s_j^-)$$

and

$$\phi(s_i^+, s_j^-) = \begin{cases} 1 & s_i^+ > s_j^- \\ 0.5 & s_i^+ = s_j^- \\ 0 & s_i^+ < s_j^- \end{cases}$$

and s_i^+ is the score for i^{th} randomly chosen positive sample and s_j^- is the score for j^{th} randomly chosen negative sample. AUC_1 and AUC_2 correspond to the ROCs of two different classifiers. To perform the DeLong test, two classifiers are trained and their predicted probabilities on the test data are obtained. The ROC curves and corresponding AUC values are computed, and the

variance of the difference between the two AUCs is estimated using the DeLong method. The test statistic is then computed as

$$Z = \frac{AUC_1 - AUC_2}{\sqrt{Var(AUC_1 - AUC_2)}}$$

which asymptotically follows a standard normal distribution, allowing computation of a p-value for testing the null hypothesis.

Generally, greater AUC indicates a better model and the hypothesis of equal prediction is rejected under small p-value.

2.3 Kolmogorov-Smirnov Test

The Kolmogorov-Smirnov test is a nonparametric test for comparing empirical distribution functions (Massey, 1951). In machine learning applications, the KS test is often used to compare score distributions, prediction distributions, or feature importance distributions across different population groups, for example in fairness analysis or model validation. There are two versions of KS test:

- **One-sample KS test:** compare sample and theoretical distribution
- **Two-sample KS test:** compare two samples

The null hypothesis for KS test is:

$$H_0 : F_1(x) = F_2(x) \forall x$$

where F_1 and F_2 are distributions for two different sample, which are defined as:

$$F_n(x) = \frac{1}{n} \sum_{i=1}^n \mathbb{I}(X_i \leq x)$$

Define

$$F_1(x) := \text{sample CDF of sample 1}$$

$$F_2(x) := \text{sample CDF of sample 2}$$

The test statistic of KS test is

$$D = \sup_x |F_1(x) - F_2(x)|$$

If the KS statistic D is sufficiently large, the null hypothesis is rejected, indicating that the two samples come from different distributions. The statistical significance of the test is determined using the asymptotic distribution of the KS statistic.

2.4 T-test and χ^2 -test

The t-test is used to determine whether the mean of a sample differs significantly from a reference value or whether the means of two samples are significantly different. The test statistic is defined as

$$t = \frac{\bar{X} - \mu}{s/\sqrt{n}}$$

which follows a Student's t-distribution under the null hypothesis.

The χ^2 test is used to determine whether there is a significant association between categorical variables or whether observed frequencies differ from expected frequencies. The test statistic is defined as

$$\chi^2 = \sum_i \frac{(O_i - E_i)^2}{E_i}$$

where O_i and E_i are observed and expected frequencies (counts) for categories, respectively.

2.5 Gini Coefficients

Gini coefficients measure the concentration or inequality of a distribution. In this paper, they are used to evaluate how unevenly a feature's importance is distributed across population groups. If the importance of a feature is concentrated in a specific group, the model may exhibit potential bias. If the importance is evenly distributed across groups, the model is considered to have lower fairness risk. Gini coefficient is defined as

$$G = 1 - \sum_{i=1}^n (F_i + F_{i-1})p_i = \frac{\sum_{i=1}^n \sum_{j=1}^n |x_i - x_j|}{2n^2 \bar{x}}$$

where

- F_i : Cumulative proportion of total value up to group i
- F_{i-1} Previous cumulative proportion
- p_i : Proportion of population in group i
- x_i : Value for individual i (e.g., income, Shapley value, importance)
- x_j : Value for individual j
- n : Number of individuals / groups
- $\sum_{i=1}^n \sum_{j=1}^n |x_i - x_j|$: Sum of difference across all pairs

2.6 Shapley Values

The Shapley value originates from cooperative game theory (Shapley, 1953) used to fairly distribute the total value of a game among players based on their contributions. Mathematically, it is defined as

$$\varphi_i[v] = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!(n-|S|-1)!}{n!} [v(S \cup \{i\}) - v(S)]$$

where

- φ_i : the Shapley value of game v for player i , which is an n -vector
- v : Value of coalition T of an n -person game
- S : A coalition
- n : Number of players
- N : Set of all players
- $v(S \cup \{i\}) - v(S)$: Marginal contribution of player i
- $\frac{|S|!(n-|S|-1)!}{n!}$: Weight corresponding to the number of possible coalition orders.

In the context of machine learning, the parameters can be interpreted as the following:

- φ_i : Feature contribution to prediction.
- v : Model prediction using features in S
- S : Subset of features

- n : Number of features
- N : Set of all features
- $v(S \cup \{i\}) - v(S)$: Change in prediction when adding feature

However, the computation of Shapley values grows exponentially with the number of players or features because the values of all possible coalitions have to be computed. Therefore, numerical approximation is often used in practice (Rozemberczki et al., 2022).

3 Introduction to KAIRI-SAFE Framework

KAIRI (Key Artificial Intelligence Risk Indicators), proposed by Giudici, Centurelli, and Turchetta (2024), is a series of indicators that aims to help organizations measure, manage, and mitigate risks throughout the AI development and deployment cycle.

3.1 Sustainability (S)

Sustainability of AI models depends on its robustness under stressed situation, which can range from extreme data input to resilience against cyber-attacks such as prompt injection. Giudici et al. proposed to use two sets of different measurements of sustainability depending on whether the model is probabilistic or non-probabilistic.

For probabilistic models (e.g. regressions and GARCH), nested models are compared and F-test (for continuous models) or χ^2 test (for binary models) is used to decide whether removing a variable significantly reduces model quality. The test is completed once removing variables shows performance would significantly worsen.

For non-probabilistic models (e.g. decision trees or neural networks), variables are ranked using explainability measure (see section 3.4). The ranked variables are then added one-by-one based on their importance. After that, the Diebold–Mariano test for continuous responses or the DeLong test for binary responses is applied. Similarly, the test is completed once prediction accuracy stops improving statistically significantly.

3.2 Accuracy (A)

Accuracy is one of the most common measures in AI performance. It measures how well a model predicts the target variable. Two different measures are used to measure accuracy for continuous and discrete models. For continuous models, rooted mean squared error (RMSE) is used:

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

where y_i is the observed value and $\hat{y}_i$ is the predicted value. Lower RMSE indicates better prediction accuracy. To compare model accuracy, Diebold-Mariano test described in 2.1 is used with null hypothesis:

$$H_0 : \text{Models have the same predictive accuracy}$$

For discrete model, predictive accuracy is commonly evaluated using the Receiver Operating Characteristic (ROC) curve and the Area Under the Curve (AUC) as described in 2.2. The test being used to compare classifier is DeLong test (see 2.2) with null hypothesis:

$$H_0 : AUC_1 = AUC_2$$

3.3 Fairness (F)

Fairness measures the bias among different population group and, optimally, should be measured at both data and prediction level. Specifically,

1. The model should not systematically disadvantage a group
2. Model predictions or feature importance should be similar across groups
3. No bias toward demographic group (gender, country, sector, etc.)

In order to measure the fairness, the Shapley value is calculated for each observation and value of the model (see section 2.6). Then the average Shapley value per group v_{mk} is calculated:

$$v_{mk} = \frac{1}{n_m} \sum_{i \in m} \phi_{ik}$$

Next, we calculate the total importance of feature k by:

$$v_k = \sum_m v_{mk}$$

After that, we calculate the importance share per group:

$$q_{mk} = \frac{v_{mk}}{v_k}$$

which is the proportion of feature importance coming from group m . This should give the following feature matrix:

$$\begin{bmatrix} q_{11} & q_{12} & \cdots & q_{1n} \\ q_{21} & q_{22} & \cdots & q_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ q_{m1} & q_{m2} & \cdots & q_{mn} \end{bmatrix}$$

where q_{mk} is the importance share of feature k attributed to group m . Finally, we calculate the Gini coefficient to the distribution q_{mk} by:

$$G_k = \frac{\sum_{m=1}^M \sum_{l=1}^M |q_{mk} - q_{lk}|}{2M}$$

Note that the term $\sum_{m=1}^M q_{mk}$ disappears because it is 1 by construction. G_k is the average difference in feature importance shares across groups. With the goal of measuring fairness of a machine learning model, G_k measures how evenly the importance of feature k is distributed across groups. It can be interpreted as following:

G_k	Interpretation
Low	Importance evenly distributed across groups; lower fairness risk
Medium	Some differences across groups
High	Importance concentrated in few groups; potential bias

A higher value of G_k indicates greater concentration of feature importance across groups, while a lower value indicates a more even distribution. Therefore, in this formulation, lower G_k values correspond to greater fairness.

3.4 Explainability (E)

Explainability refers to the ability to understand and interpret the contribution of each feature to the model predictions. In the context of AI risk management, explainability is essential to ensure transparency, accountability, and regulatory compliance.

In the past several years, there have been a large number of new approaches developed to visualize the mechanism of AI models. However, they are all limited in the sense that they are model-specific. For example, GRAD CAM is primarily focused on visual neural networks, and causal tracing is specifically designed to trace BERT-based model such as GPT. This poses a challenge for enterprise to develop AI explainability measure that is comparable across different model families.

Similar to fairness in section 3.3, explainability is measured using Shapley values, which quantify the contribution of each feature to the model prediction. However, instead of Gini coefficient, t-test is applied to compare model explainability.

Let ϕ_{ik} denote the Shapley value of feature k for observation i . The global importance of feature k is defined as

$$\Phi_k = \frac{1}{n} \sum_{i=1}^n \phi_{ik}$$

To assess whether a feature contributes significantly to the model, a t-test is performed on the Shapley values. The t-test is used because explainability is evaluated through the mean contribution of features, and the test determines whether these contributions are statistically different from zero. The null hypothesis is

$$H_0 : \Phi_k = 0$$

with test statistic:

$$t_k = \frac{\Phi_k}{s_k / \sqrt{n}}$$

where s_k is the standard deviation of the Shapley values for feature k .

If the null hypothesis is rejected, the feature is considered to have a statistically significant contribution to the model prediction. Otherwise, the feature may be considered irrelevant or unstable.

4 Mapping from AI Risks to Financial Risks

In the original paper of Giudici et al., credit risk model was used as an example for the financial application of KAIRI-SAFE framework. However, we will attempt to generalize the existing framework to more general financial risk management framework.

Traditional model risk management emphasizes model development, validation, implementation, and ongoing monitoring (Board of Governors of the Federal Reserve System and Office of the Comptroller of the Currency, 2011):

1. **Risk appetite definition:** Determining the maximum level of risk an institution is willing to accept in alignment with its strategic objectives.
2. **Risk identification and mapping:** Identifying sources of risk, assessing their potential impact, and determining appropriate management strategies.
3. **Operationalization of risk appetite:** Translating risk appetite into measurable indicators, defining risk limits, and allocating appropriate resources, expertise, and infrastructure.
4. **Implementation:** Deploying models, strategies, and controls within operational systems.

5. **Monitoring and re-evaluation:** Continuously assessing model performance and risk exposure over time.

Furthermore, financial institutions typically adopt four types of risk management strategies:

1. **Risk retention:** Accepting the risk when it falls within the defined risk appetite.
2. **Risk avoidance:** Avoiding activities that exceed acceptable risk levels.
3. **Risk mitigation:** Reducing risk through controls, model adjustments, or diversification strategies.
4. **Risk transfer:** Transferring risk through financial instruments such as insurance or derivatives.

Each component of the SAFE framework can be translated into measurable indicators that correspond to specific types of financial risk. These indicators allow AI-related risks to be quantified and monitored within a traditional risk management framework.

In particular, the SAFE metrics are associated with different statistical quantities and corresponding hypothesis tests, which enable the construction of operational risk indicators. These indicators can then be linked to financial risk categories.

SAFE	Indicator	Statistical Tool	Financial Risk
Sustainability	Accuracy Change	DM / DeLong	Model Risk
Accuracy	RMSE / AUC	DM / DeLong	Market / Credit Risk
Fairness	G_k and KS	Gini and KS	Regulatory Risk
Explainability	Φ_k and T-test	T-test	Governance Risk

Table 1: Mapping from SAFE components to financial risks

The indicators derived from SAFE metrics provide a quantitative basis for evaluating AI systems. For example, statistically significant changes in predictive accuracy may indicate model instability, while deviations in fairness metrics such as the Gini coefficient or KS statistic may signal potential bias. Similarly, insignificant Shapley values may indicate a lack of explainability.

By expressing SAFE metrics as measurable indicators, AI risks can be systematically integrated into financial risk management processes.

4.1 Framework Mapping

The SAFE indicators can be integrated into each stage of the traditional risk management process, allowing AI risks to be managed in a structured and operational manner.

Step 1: Risk Appetite Definition

Risk appetite can be expressed in terms of acceptable thresholds for SAFE indicators. For example, minimum acceptable predictive performance, fairness thresholds, and statistical significance levels can be defined to determine whether a model meets the institution’s requirements. If the model is utilized for high-stakes operation (for example with large amount of capital involved or for highly volatile derivatives), a higher confidence level should be used.

For example, an institution may require:

1. Minimum predictive performance (e.g. $AUC > 0.75$ or RMSE below a predefined level)
2. Minimum fairness score (e.g. $F_k > 0.6$) for $F_k = 1 - G_k$

3. Statistical significance of feature contributions (e.g., p-value < 0.05 for key features)

In this way, SAFE indicators provide a quantitative representation of risk appetite, allowing institutions to explicitly define acceptable levels of AI risk.

Step 2: Risk Identification and Mapping

SAFE metrics are used to identify and classify different types of AI-related risk. Each SAFE component corresponds to a distinct risk category:

SAFE metrics	Risk Identification
Sustainability	Model instability and overfitting
Accuracy	Financial loss due to prediction error
Fairness	Bias and regulatory risk
Explainability	Governance and transparency risk

Table 2: SAFE components and corresponding AI risk categories

For example:

1. A model with unstable performance across different samples may indicate high model risk.
2. Significant differences in feature importance across groups (high G_k or low transformed fairness score F_k) may indicate potential bias.
3. Insignificant Shapley values may suggest that the model relies on noise rather than meaningful signals.

This step transforms SAFE metrics into diagnostic tools for identifying risk sources.

Step 3: Operationalization of Risk Appetite

In this stage, SAFE metrics are converted into operational risk indicators that can be monitored and acted upon. Each metric becomes an indicator with associated thresholds and triggers:

Indicator	Signal	Risk Interpretation
RMSE increase	Performance deterioration	Accuracy Risk
AUC decrease	Reduced classification quality	Financial Model Risk
High G_k or low F_k	Uneven importance distribution	Fairness Risk
Significant KS statistic	Distribution difference	Bias Risk
Insignificant t-test	Weak feature contribution	Explainability Risk

For example:

1. A statistically significant increase in RMSE may trigger model review.
2. An increase in G_k above the threshold, or equivalently a drop in the transformed fairness score F_k below the threshold, may trigger internal audit to determine whether the uneven feature weight distribution is justified.
3. Significant KS test results may indicate the need for rebalancing data or features.

These indicators allow institutions to define risk limits and establish automated monitoring rules, similar to traditional financial risk indicators.

Step 4: Implementation

SAFE metrics may be used as a guide for model selection and validation. Models that meet predefined thresholds for sustainability, accuracy, fairness, and explainability are deployed, while those that do not meet these criteria are rejected or modified.

Step 5: Monitoring and Re-evaluation

SAFE indicators are continuously monitored over time to detect model drift, performance degradation, and emerging bias. Changes in RMSE, AUC, Gini coefficients, or Shapley significance can signal increasing risk and trigger re-evaluation or retraining of models.

4.2 Alignment with Existing AI and Model Risk Frameworks

The proposed SAFE-based risk management process is consistent with existing AI risk and financial model risk management frameworks. In particular, the NIST AI Risk Management Framework organizes AI risk management around four core functions: Govern, Map, Measure, and Manage (National Institute of Standards and Technology, 2023). The SAFE framework can be mapped naturally to these functions. The risk appetite and governance steps correspond to the Govern function, the identification of AI-related financial risks corresponds to the Map function, the statistical evaluation of SAFE indicators corresponds to the Measure function, and the use of thresholds, monitoring, and remediation actions corresponds to the Manage function. The framework is also consistent with ISO/IEC 23894:2023, which provides guidance for organizations developing, deploying, or using AI systems to manage AI-related risks and integrate risk management into AI-related activities (International Organization for Standardization, 2023). In this paper, SAFE indicators serve as measurable risk controls that allow AI-specific risks, such as instability, bias, poor predictive performance, and lack of explainability, to be incorporated into an institution’s broader risk management process.

From a financial regulatory perspective, the proposed framework is aligned with model risk management principles such as those described in SR 11-7, which emphasizes model development, validation, implementation, governance, controls, and ongoing monitoring (Board of Governors of the Federal Reserve System and Office of the Comptroller of the Currency, 2011). The SAFE framework extends these principles to AI models by defining quantitative indicators for model robustness, predictive performance, fairness, and explainability. In this sense, SAFE indicators can support model validation and ongoing monitoring by providing diagnostic signals for different dimensions of AI model risk.

The framework is also related to Basel operational risk and risk reporting principles. The Basel Committee emphasizes governance, control environments, information systems, business continuity, and risk reporting as part of sound operational risk management (Basel Committee on Banking Supervision, 2021). Similarly, BCBS 239 emphasizes effective risk data aggregation and risk reporting (Basel Committee on Banking Supervision, 2013). The proposed SAFE indicators can contribute to these objectives by transforming AI model characteristics into measurable indicators that can be monitored, aggregated, and reported to risk committees or senior management.

Finally, the framework is compatible with the risk-based approach of the EU AI Act, which imposes stronger requirements on high-risk AI systems (European Parliament and Council of the European Union, 2024). In financial applications such as credit scoring, insurance underwriting, fraud detection, and risk forecasting, AI models may have material consequences for individuals and institutions. The SAFE framework supports this risk-based approach by requiring stricter thresholds, stronger explainability, and more intensive monitoring for models used in high-stakes applications.

Overall, the proposed framework does not replace existing AI or financial risk management

standards. Rather, it provides a quantitative layer that helps operationalize them. By translating sustainability, accuracy, fairness, and explainability into measurable indicators, the SAFE framework connects high-level governance principles with practical model validation, monitoring, and remediation procedures.

4.3 Sustainability and Model Risk

Sustainability is closely related to model risk, which arises when a model produces unreliable or unstable predictions. Overly complex models may overfit the training data and fail under stressed or out-of-sample conditions. The SAFE framework addresses this by promoting parsimonious models and testing whether additional variables significantly improve predictive performance using statistical tests such as the Diebold–Mariano and DeLong tests. By selecting models that are both accurate and stable, sustainability reduces model risk and enhances robustness.

For sustainability, the Diebold–Mariano test and the DeLong test provide p-values. However, to set a specific critical value, it is crucial to understand:

1. Type of financial product being modeled
2. Volatility and liquidity of the underlying assets
3. Amount of required economic and regulatory capital

Higher-risk products require stricter SAFE thresholds and stronger validation requirements.

4.4 Accuracy and Market/Credit Risk

Accuracy is directly related to financial risk because inaccurate predictions can lead to incorrect pricing, portfolio allocation, credit decisions, and risk estimation.

For continuous prediction models, accuracy is commonly measured using RMSE. In financial applications, these models may be used to forecast asset prices, portfolio returns, volatility, or Value-at-Risk (VaR). High RMSE indicates that the model predictions deviate significantly from observed market behavior, increasing the probability of incorrect investment decisions, mispricing, or underestimation of financial risk. In particular, inaccurate volatility or tail-risk estimation may lead to insufficient capital allocation and excessive exposure during stressed market conditions.

For binary classification models, predictive accuracy is typically measured using ROC curves and AUC. These models are widely used in credit scoring, fraud detection, anti-money laundering systems, and bankruptcy prediction. A low AUC indicates poor discriminatory power between positive and negative outcomes, increasing the probability of incorrect credit decisions or failure to detect fraudulent behavior. In credit risk applications, this may result in underestimation of default probabilities, increased non-performing loans, and inadequate regulatory capital allocation.

To determine whether differences in predictive performance are statistically significant, the SAFE framework employs the Diebold–Mariano test for continuous models and the DeLong test for binary classifiers. These tests allow institutions to distinguish between genuine improvements in model performance and random variation, reducing the risk of selecting unstable or overfitted models.

At the operational level, the indicators can be interpreted as the following:

Indicator	Interpretation	Risk Signal
RMSE increase	Prediction deterioration	Market risk
AUC decrease	Reduced classification quality	Credit risk
Significant DM test	Model instability	Market risk
Significant DeLong test	Material performance difference	Validation issue

Table 3: Operational interpretation of accuracy indicators

Accuracy indicators should be monitored continuously throughout the model lifecycle. Significant increases in RMSE or decreases in AUC may indicate model drift, changing market conditions, or degradation in predictive quality. When predefined thresholds are breached, institutions may respond through model recalibration, retraining, or replacement.

4.5 Fairness and Regulatory Risk

Fairness is closely related to regulatory, legal, and reputational risk in financial applications. AI models that systematically disadvantage specific demographic or economic groups through biased predictions may violate regulatory standards, expose institutions to litigation, and damage stakeholder trust. In addition, biased model behavior may indicate instability or overreliance on group-specific patterns, reducing the robustness and generalizability of the model.

Fairness is particularly important in applications such as credit scoring, insurance underwriting, fraud detection, and portfolio recommendation systems, where model outputs directly influence financial access and economic outcomes. Unequal treatment across demographic groups may result in discriminatory lending practices, unequal pricing, or exclusion from financial services. Regulatory frameworks increasingly require financial institutions to demonstrate that AI systems do not produce discriminatory outcomes. Biased models may violate anti-discrimination regulations, consumer protection standards, and emerging AI governance frameworks such as the EU AI Act and NIST AI RMF. As a result, fairness has become both an ethical and regulatory requirement for AI deployment in finance.

Within the SAFE framework, fairness is evaluated by examining how feature importance is distributed across population groups. A high value of G_k , or equivalently a low transformed fairness score F_k , indicates that feature importance is concentrated within specific groups, suggesting that the model relies disproportionately on group-specific characteristics.

The Kolmogorov–Smirnov test is used to compare prediction score distributions across groups. Significant KS statistics indicate that the model generates systematically different prediction distributions for different populations. Such differences may signal unequal treatment, data imbalance, or unintended discrimination.

Fairness can also be evaluated through predictive performance across groups. Significant differences in AUC values may indicate that the model performs substantially better for certain groups than others. In credit risk applications, this may result in systematically inaccurate risk assessments for disadvantaged populations.

Fairness indicators should be monitored continuously throughout the model lifecycle. Declining fairness metrics may indicate changes in data distributions, emerging demographic bias, or unintended model behavior. However, fairness indicators should not be interpreted mechanically. A deterioration in fairness metrics may indicate potential bias, but it does not necessarily imply discriminatory behavior. The economic and institutional context of the modeled object must also be considered. For instance, when modeling global counterparty risk using machine learning techniques, the model may assign substantial importance to country-specific variables. This does not necessarily mean that the model discriminates against counterparties from a given country; rather, it may indicate that country-level characteristics, such as sovereign risk, legal enforceability, macroeconomic conditions, capital controls, trade restrictions, or geopolitical conditions, have a genuine effect on counterparty risk in global trade.

Therefore, fairness indicators should be interpreted as diagnostic signals rather than automatic evidence of discrimination, and any detected imbalance should be reviewed together with domain experts. When predefined thresholds are breached and the imbalance is determined to be problematic, institutions may implement corrective actions such as data rebalancing, feature adjustment, fairness constraints, or model retraining.

4.6 Explainability and Governance Risk

Explainability is closely related to governance, compliance, and operational risk in financial applications. Financial institutions are often required to justify model outputs to internal stakeholders, auditors, regulators, and affected customers. When an AI model is highly complex or opaque, it may be difficult to determine why a specific prediction or decision was made. This creates governance risk because the institution may be unable to validate, audit, or defend the model’s behavior.

This issue is particularly important for applications such as credit scoring, fraud detection, portfolio recommendation, and risk forecasting, where model outputs can directly affect lending decisions, investment allocation, capital requirements, or customer treatment. A model with high predictive accuracy but low explainability may still be unsuitable for deployment if its decisions cannot be interpreted or justified.

Within the SAFE framework, explainability is evaluated using Shapley values, which quantify the contribution of each feature to model predictions. The global importance of feature k is given by Φ_k , and the statistical significance of this contribution can be assessed using a t-test. If the null hypothesis $H_0 : \Phi_k = 0$ is rejected, the feature is considered to have a statistically significant contribution to model prediction. Conversely, insignificant Shapley values may indicate that the model relies on unstable or weak explanatory signals.

Explainability indicators can therefore be used as governance risk indicators. For example, if only a small number of features have statistically significant contributions, the model may lack a clear explanatory structure. Similarly, if feature importance changes substantially over time, this may indicate instability in the model’s decision logic. Such behavior can create difficulties for model validation, internal audit, regulatory review, and business oversight.

However, explainability should also be interpreted in context. A complex model is not necessarily inappropriate simply because it is difficult to interpret; rather, the required level of explainability depends on the use case, materiality, and potential impact of the model. For high-stakes applications such as credit approval, capital allocation, or derivative pricing, stronger explainability requirements should be imposed. For lower-risk applications, partial explainability may be acceptable if the model satisfies other SAFE requirements.

When explainability indicators fall below predefined thresholds, institutions may implement corrective actions such as simplifying the model, reducing the number of features, replacing black-box models with more interpretable alternatives, producing model documentation, or requiring additional human oversight. In this way, explainability metrics support model governance by ensuring that AI systems remain transparent, auditable, and accountable throughout their lifecycle.

In financial applications, explainability is not only a technical property of a model, but also a governance requirement because institutions must be able to justify, audit, and monitor model-driven decisions.

Indicator	Signal	Risk Interpretation
Low number of significant Shapley values	Weak explanatory structure	Governance risk
Unstable feature importance over time	Changing decision logic	Model validation risk
High model complexity	Limited interpretability	Auditability risk
Insignificant t-test results	Weak feature contribution	Explainability risk

Table 4: Operational interpretation of explainability indicators

5 Aggregate SAFE Rating for AI Model Assessment

After individual SAFE indicators are computed, financial institutions require a method to aggregate them into an overall assessment of model acceptability. Since sustainability, accuracy, fairness, and explainability are measured using different statistical quantities, the raw indicators must first be transformed into normalized scores on a common scale. The purpose of the aggregate SAFE rating is to translate multiple statistical indicators into a single model-level risk assessment while preserving the principle that severe weaknesses in one dimension should not be mechanically offset by strength in another.

Figure 1 summarizes the proposed integration of SAFE indicators into financial model risk management. The following subsections describe each stage of this aggregation process in detail.

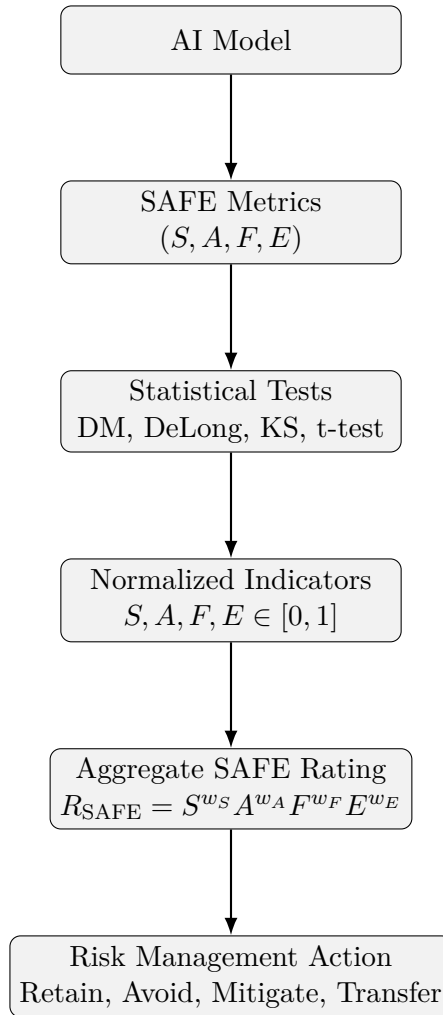


Figure 1: Integration of SAFE indicators into financial model risk management.

5.1 Normalization of SAFE Indicators

Because SAFE indicators are measured on different scales, each metric must be transformed into a normalized score in $[0, 1]$ before aggregation. Let

$$S, A, F, E \in [0, 1]$$

denote the normalized scores for sustainability, accuracy, fairness, and explainability, respectively, where higher values indicate lower AI risk. If any component score is zero, the aggregate rating becomes zero. This reflects the conservative assumption that a complete failure in one SAFE dimension represents a critical model risk.

For a metric where higher values indicate better performance, a normalized score may be defined as

$$X^* = \min \left\{ 1, \frac{X}{\tau_X} \right\},$$

where τ_X is the minimum acceptable threshold.

For a metric where lower values indicate better performance, such as RMSE, a normalized score may be defined as

$$X^* = \max \left\{ 0, 1 - \frac{X - \tau_X}{X_{\max} - \tau_X} \right\}.$$

The normalization procedure depends on the institution's risk appetite, regulatory requirements, and historical model performance. For metrics where larger values indicate better performance, such as AUC or fairness scores, normalization may be performed relative to predefined thresholds. For metrics where lower values indicate better performance, such as RMSE, the transformation should be reversed so that higher normalized values always indicate lower risk.

For example, AUC may be transformed into an accuracy score, while the raw concentration measure G_k may be transformed into a fairness score using $F_k = 1 - \tilde{G}_k$, where $\tilde{G}_k$ is the normalized concentration score. Similarly, RMSE may be transformed into an accuracy score by comparing it with an institution-defined tolerance level or historical benchmark. The main requirement is that all transformed SAFE scores follow the same interpretation: values closer to one indicate lower risk, while values closer to zero indicate higher risk.

5.2 Nonlinear Aggregation

A simple additive score may fail to capture critical weaknesses because high performance in one dimension can offset poor performance in another. For example, a model with high predictive accuracy but severe unfairness or low explainability may still receive an acceptable score under a linear weighted average. This is undesirable in high-stakes financial applications, where certain risk dimensions may be non-compensable.

Therefore, we propose a nonlinear aggregation using the weighted geometric mean:

$$R_{\text{SAFE}} = S^{w_S} A^{w_A} F^{w_F} E^{w_E}$$

where

$$w_S + w_A + w_F + w_E = 1.$$

Weighted geometric aggregation is commonly used in multi-criteria decision analysis when poor performance in one dimension should not be fully compensated by strong performance in another (Ishizaka and Nemery, 2013).

The weights w_S, w_A, w_F, w_E reflect the relative importance of each SAFE component and should be determined according to model use case, materiality, and regulatory exposure. For example, accuracy may receive greater weight in trading models, fairness may receive greater weight in credit scoring or insurance underwriting, and explainability may receive greater weight in models subject to regulatory review.

The weighted geometric mean penalizes low component scores more strongly than a linear weighted average. If one SAFE component is close to zero, the overall rating decreases significantly, reflecting the idea that a severe weakness in one risk dimension should materially affect the model-level assessment.

The proposed aggregation should be interpreted as a governance tool rather than a universal mathematical measure of AI risk. The choice of normalization method, weights, and thresholds should be calibrated to the specific use case, institutional risk appetite, and regulatory context.

5.3 Threshold Constraints and Risk Actions

In addition to nonlinear aggregation, institutions may impose minimum threshold constraints for each SAFE component. Let $\tau_S, \tau_A, \tau_F, \tau_E$ denote the minimum acceptable scores for sustainability, accuracy, fairness, and explainability. The final AI model rating can then be expressed as

$$R_{\text{AI}} = \begin{cases} S^{w_S} A^{w_A} F^{w_F} E^{w_E}, & \text{if } S \geq \tau_S, A \geq \tau_A, F \geq \tau_F, E \geq \tau_E, \\ \text{Reject or escalate,} & \text{otherwise.} \end{cases}$$

This threshold rule prevents critical weaknesses from being hidden by strong performance in other dimensions. For instance, a model with high accuracy but extremely low fairness should not be automatically approved simply because its aggregate score is acceptable. Instead, violation of a minimum threshold should trigger rejection, escalation, or additional model controls. The aggregate SAFE rating can be mapped to risk management actions as follows:

SAFE Rating	Interpretation	Risk Management Action
High	Model satisfies SAFE requirements	Retain / Deploy
Medium	Model acceptable with weaknesses	Mitigate / Add controls
Low	Model fails important requirements	Retrain / Modify
Critical breach	One or more thresholds violated	Reject / Escalate

Table 5: Mapping aggregate SAFE ratings to risk management actions

The aggregate SAFE rating provides a structured way to translate multiple statistical indicators into a model-level risk assessment, while preserving the principle that critical weaknesses in fairness, explainability, sustainability, or accuracy should not be ignored.

This aggregation framework completes the translation of SAFE indicators from individual statistical diagnostics into an institution-level model risk assessment tool.

5.4 Illustrative Application: Credit Scoring Model Assessment

To illustrate how the aggregate SAFE rating may be used in practice, consider a financial institution evaluating a machine learning model for consumer credit scoring. The model is used to predict whether a loan applicant will default within the next twelve months. Because the model affects credit approval decisions, it is subject to accuracy, fairness, explainability, and governance requirements.

Suppose the institution evaluates the model using the four SAFE dimensions and obtains the following normalized scores:

SAFE Component	Metric Used	Result	Normalized Score
Sustainability	Stress performance stability	Moderate drift detected	$S = 0.80$
Accuracy	AUC on validation data	$AUC = 0.78$	$A = 0.85$
Fairness	Transformed Gini fairness score	Some group imbalance	$F = 0.62$
Explainability	Significant Shapley contributions	Acceptable but limited	$E = 0.70$

Table 6: Illustrative SAFE assessment for a credit scoring model

Assume that the institution assigns equal weights to the four SAFE dimensions:

$$w_S = w_A = w_F = w_E = 0.25.$$

The aggregate SAFE rating is then calculated as

$$R_{\text{SAFE}} = S^{w_S} A^{w_A} F^{w_F} E^{w_E}.$$

Substituting the normalized scores gives

$$R_{\text{SAFE}} = 0.80^{0.25} \times 0.85^{0.25} \times 0.62^{0.25} \times 0.70^{0.25} \approx 0.74.$$

The institution may map the aggregate SAFE rating to model risk management actions as follows:

SAFE Rating Range	Risk Management Action
$R_{\text{SAFE}} \geq 0.80$	Retain / Deploy
$0.65 \leq R_{\text{SAFE}} < 0.80$	Mitigate / Add controls
$0.50 \leq R_{\text{SAFE}} < 0.65$	Retrain / Modify
$R_{\text{SAFE}} < 0.50$	Reject / Escalate

Table 7: Illustrative mapping from aggregate SAFE rating to risk management action

Since the model has an aggregate SAFE rating of approximately 0.74, it falls into the “Mitigate / Add controls” category. This indicates that the model may be acceptable for conditional deployment, but additional safeguards are required before full operational use.

The main weakness in this example is the fairness score, $F = 0.62$, which is lower than the other SAFE components. This suggests that the model may rely unevenly on certain features across population groups. However, the fairness result should be interpreted as a diagnostic signal rather than automatic evidence of discriminatory behavior. A model validation team should investigate whether the observed imbalance is economically justified, statistically stable, and consistent with applicable regulatory requirements.

In addition to the aggregate rating, the institution may impose minimum component-level thresholds. For example, suppose the minimum acceptable score for each SAFE component is

$$S, A, F, E \geq 0.60.$$

In this case, the model satisfies all minimum component-level thresholds, but the fairness score is close to the lower bound. Therefore, the appropriate decision is conditional deployment with fairness mitigation and enhanced monitoring. Possible controls include group-level performance monitoring, periodic recalibration, additional documentation of economically justified variables, and review by model risk management or internal audit. This example demonstrates the purpose of nonlinear SAFE aggregation. Although the model has acceptable accuracy, its weaker fairness score materially reduces the overall model rating. The weighted geometric mean therefore prevents strong performance in accuracy from fully offsetting weaknesses in other dimensions. In this way, the aggregate SAFE rating provides a structured mechanism for translating statistical model diagnostics into financial model risk management decisions.

6 Conclusion

The increasing adoption of artificial intelligence in financial services creates new challenges for model risk management. While machine learning models may improve predictive performance in areas such as credit scoring, portfolio management, fraud detection, and risk forecasting, they also introduce risks related to robustness, accuracy, fairness, and explainability. These risks are difficult to manage using traditional model validation tools alone, particularly when AI systems are complex, opaque, and continuously updated.

This paper builds on the KAIRI-SAFE framework by examining how Sustainability, Accuracy, Fairness, and Explainability can be measured using statistical indicators and integrated into financial risk management processes. The Diebold–Mariano test, DeLong test, Kolmogorov–Smirnov test, t-test, Shapley values, and Gini coefficients are introduced as tools for evaluating different dimensions of AI model risk. Each SAFE component is then mapped to a corresponding financial risk category, including model risk, market and credit risk, regulatory risk, and governance risk.

The main contribution of this paper is to translate SAFE metrics into operational risk indicators that can be incorporated into a traditional risk management cycle. By defining thresholds, monitoring indicators over time, and linking metric breaches to risk management actions, the proposed framework allows AI risks to be evaluated in a manner familiar to financial institutions. In addition, the aggregate SAFE rating provides a structured method for combining multiple AI risk indicators into a model-level assessment, while ensuring that severe weaknesses in one dimension are not offset mechanically by strength in another. The illustrative credit scoring example further shows how normalized SAFE indicators can be aggregated into a model-level rating and linked to conditional deployment, mitigation, retraining, or escalation decisions.

Several limitations remain. First, the choice of thresholds and weights in the aggregate SAFE rating depends on institutional risk appetite, regulatory expectations, and the specific financial application. Second, fairness and explainability indicators must be interpreted in context, as statistical imbalance does not necessarily imply discriminatory behavior or model failure. Third, the proposed framework requires empirical validation across different financial use cases and market environments. Future research may focus on calibrating SAFE thresholds using historical model performance, stress testing aggregate SAFE ratings, and extending the framework to unsupervised and generative AI systems.

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